Harsha Raj Kumar

Nashville, TN (open to relocation) | harsha.raj.kumar@vanderbilt.edu | +1 (615) 482-9742
linkedin.com/in/harsharajkumar273 | github.com/harsharajkumar-273 | Portfolio

EDUCATION

Vanderbilt University - M.S. Computer Science **Expected April 2027**
GPA: **3.6/4.0** | Nashville, TN

Vellore Institute of Technology - B.Tech. Computer Science **May 2025**
GPA: **3.5/4.0** | Chennai, India

EXPERIENCE

Amplify GenAI Innovation Center, Vanderbilt University **Aug 2026–Present**
Research Assistant, Software Engineering *React, TypeScript, Python, AWS Lambda, DynamoDB, PostgreSQL/pgvector*

- Implemented schema-validated project CRUD, file-manifest, and memory APIs in Python/AWS Lambda, backed by **three DynamoDB tables** and server-side ownership checks.
- Built the React/TypeScript project workspace and server-side chat-context integration, attaching project instructions, knowledge-base files, and approved memory to conversations.
- Cut retrieval latency **38%** using PostgreSQL/pgvector hybrid search, document chunking, and caching.
- Integration-tested Projects before release, identifying an internal-field exposure in sharing and more than **20 additional defects**.

VU-BEAM Lab, Vanderbilt University **Oct 2025–Aug 2026**
Research Assistant *Python, PyTorch, Slurm*

- Implemented contrastive pretraining and supervised fine-tuning for ultrasound research, including PyTorch data loading and augmentation.

Vanderbilt University Mathematics Department **Sep 2025–May 2026**
Research Assistant, Software Engineering - Proofdesk Live *React, TypeScript, Node.js/Express, Y.js, Redis/BullMQ, Docker, Pyodide*

- Co-developed Proofdesk with Y.js/WebSocket collaboration for multiple users; enabled a professor to publish and maintain an online textbook, with an estimated **20%** improvement in update-workflow efficiency.
- Implemented Docker build execution behind Express APIs with BullMQ/Redis and **512 MB / 64-process limits**; maintained GitHub Actions CI/CD for tests, builds, Docker Compose deployments, and health checks.
- Continued maintaining Proofdesk after the appointment; added a Pyodide browser preview in June 2026 to render PreTeXt without server round-trips.

OPEN SOURCE

PyTorch | **2 merged contributions** | NCCL2 #200585 | Core #191092

- Fixed C++ NCCL2 `all_gather` misclassifying empty tensors as aliased buffers, preventing shape errors in uneven gathers; also fixed zero-dimension validation in `torch.unravel_index`, with regression tests for both.

PROJECTS

PulseStream *TypeScript, Kafka/Redpanda, PostgreSQL, Redis, Kubernetes*
github.com/harsharajkumar-273/PulseStream

- Built an Express/Kafka/PostgreSQL pipeline recording **2,933 persisted events/sec** in a **30-second local Docker benchmark**; validated **11 API-level failure scenarios** in CI.
- Fixed batch data loss with PostgreSQL savepoints and post-commit Kafka offset resolution; recovered **3,000 events** across **three local SIGKILLs**.

LSM-Tree Storage Engine *C++17, Linux io_uring*
github.com/harsharajkumar-273/lsm_tree

- Co-developed and maintain an LSM-tree engine with a CAS-based SkipList MemTable, asynchronous WAL, SSTables, blocked Bloom filters, and streaming compaction.
- Reviewed WAL ordering and recovery; maintain **14 CTest programs** covering randomized operations, crash recovery, cancellation, and injected faults with contributors.

CodeGraph *TypeScript, Tree-sitter*
github.com/harsharajkumar-273/CodeGraph

- Published codegraph-engine on npm; built dependency/call graphs and reverse-edge traversal for change impact, with automated regression tests.
- Added resolved/probable/ambiguous edge labels and source-level traceability so developers can inspect inferred dependencies before acting on change-impact results.

SKILLS

Languages: C++17, Python, TypeScript, Java, SQL
Backend/Data: Node.js/Express, FastAPI, PostgreSQL/pgvector, DynamoDB, Redis, Kafka, Apache Beam
Web/Infrastructure: React, WebSockets, Linux, Docker, Kubernetes, AWS Lambda, GitHub Actions CI/CD